Does Tessera's larger reconstruction error matter downstream?

Interim answer: above ~1 bit/pixel, no — the codec is task-transparent despite a cosine error ~10x AlphaEarth's. Below ~0.5 bit/pixel, yes — Tessera degrades roughly twice as fast. Measured on AlphaEarth's own benchmark tasks, at the same label points, with the same probe protocol.

1 Cosine error does not transfer across sources.

At matched — indeed slightly worse — reconstruction cosine, the Tessera codec costs LESS downstream accuracy than the AlphaEarth codec does. On LUCAS land cover, AlphaEarth at cos 0.953 loses 4.3 points of balanced accuracy; Tessera at cos 0.948 loses 0.5. Tessera's error budget lands in dimensions these tasks do not read.

2 Above ~1 b/px the Tessera codec is downstream-transparent.

At the top of the rate ladder every task sits within its split-noise floor — the same verdict the AlphaEarth study reached, reached at a higher rate. Tessera does this at cosine ~0.95, where AlphaEarth needs ~0.99.

3 Below ~0.5 b/px Tessera degrades faster — clearly so on the one task large enough to tell.

On LUCAS (16,780 points) the lowest-rate loss is 2x AlphaEarth's and holds across seeds: -0.108 ±0.010 against -0.052 ±0.009, a separation five times its own noise. On LCMAP (3,008 points) the two are indistinguishable — -0.054 ±0.009 against -0.054 ±0.020. Treat the low-rate penalty as established on LUCAS only.

4 On the same points, the uncompressed sources are not equal.

AlphaEarth scores higher on 4 of 5 tasks (LCMAP LC 0.818 vs 0.750); the two tie on LUCAS land cover (0.5635 vs 0.5650). Tessera's ~2.5x higher latent information density does not show up as a probe advantage here.

Significance was measured, not assumed: every rate point was re-probed under repeated geo-block split seeds (page 5). The resulting noise floor reaches ±0.017 balanced accuracy on LUCAS and ±0.023 on LCMAP— so LCMAP cannot resolve anything smaller than ~0.05, and its apparent 'compression beats the original' points are artefacts. Claims above are stated only where they clear their own noise floor.

What was measured

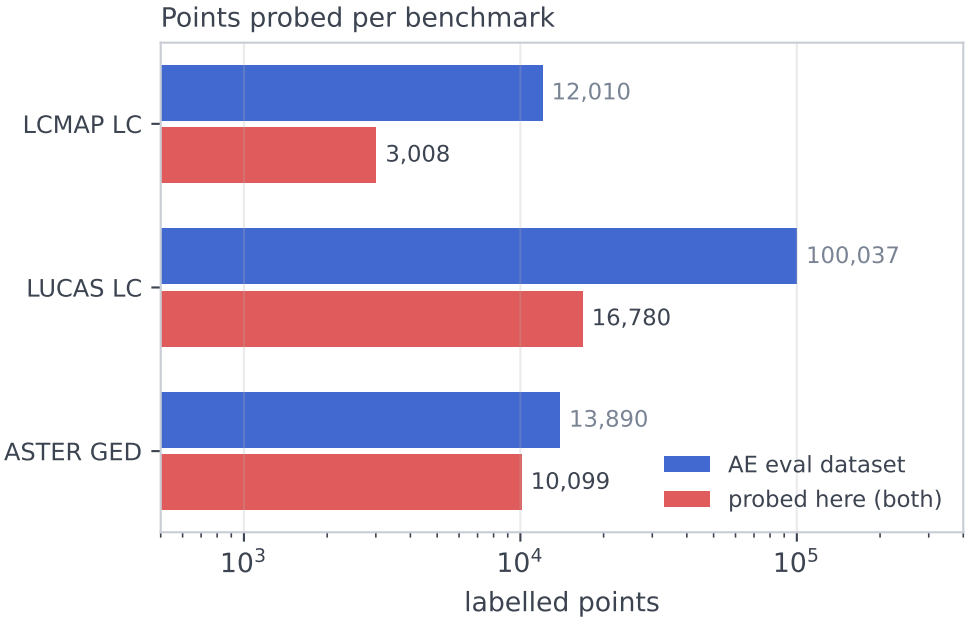
The AlphaEarth downstream study, re-run unchanged on Tessera embeddings at the same locations.

The protocol (unchanged from the AlphaEarth study, itself a reimplementation of the AlphaEarth Foundations paper's frozen-embedding probes):

- Encode each labelled pixel's 64x64 context window with the codec, decode it at a given RVQ rate tuple, and read the reconstructed embedding back at the label pixel.
- Fit probes on the reconstruction — 1-nearest-neighbour, 3-NN and a linear probe (logistic / ridge regression) — and compare against the same probes fit on the uncompressed embedding.
- Split train/test by source tile (~82 km blocks), so no tile straddles the split and spatial autocorrelation cannot leak.
- Report the metric delta (reconstruction minus original) as a function of bits per pixel. Delta, not absolute score, is the compression cost.

Matching the two sources:

- The label points come from the built AlphaEarth eval datasets. For each point we re-extracted a 128 px Tessera context crop from the 0.1-degree Tessera tile containing it, using the tile's landmask georeferencing, and the tile year nearest the AlphaEarth target year (mostly 2018).
- Any Tessera tile that contributed chips to the codec's training store is excluded, in every year — so no eval point was seen in training, on either side.
- The AlphaEarth curves shown here were re-run restricted to exactly the points Tessera reached, so both curves describe the same ground.



Tessera counts are lower because a 0.1° tile is a ~110 MB download: LUCAS and LCMAP were capped at a 2,000-tile budget (densest-first), and points on training tiles or outside Tessera coverage were dropped. ASTER used all 409 covered tiles.

Codec under test: run tessera_vq8 (wandb 5bcr0y0u) — a 3-tier pyramid RVQ, 64 px window, 8 levels per tier, validation cosine 0.962. This run is mid-training and is NOT the final Tessera codec. AlphaEarth reference: ae_pyr64_divfull, the shipped release model.

The compression cost, task by task

Δ metric (reconstruction − uncompressed) against rate. Zero is free compression; the shaded band is ±0.01, about the width of the split-noise floor. Both codecs converge to it — Tessera later.



Reading these panels

The gap between the red and blue curves at the left-hand edge is the real cost of Tessera's noisier field: at 0.07 b/px the Tessera codec gives up 2-4x more accuracy than the AlphaEarth codec does.

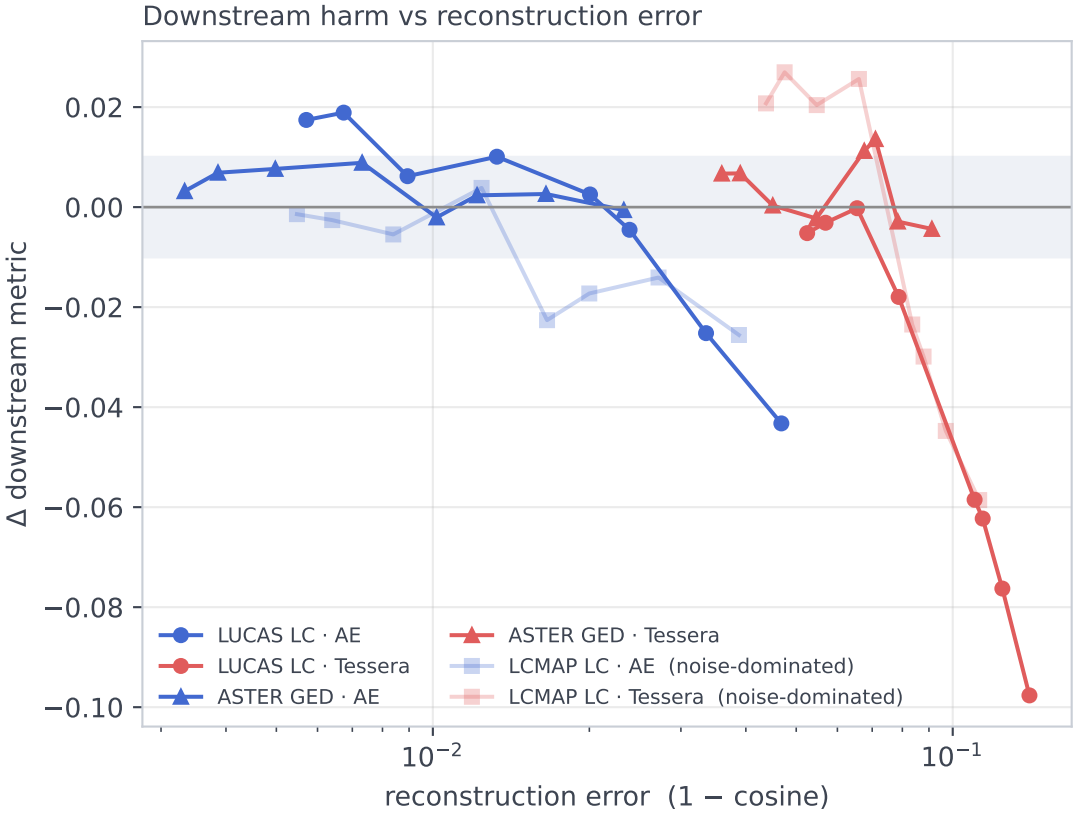
By 1-2 b/px the curves have merged into the noise band and the source stops mattering for compression cost.

Note the vertical scales differ: ASTER's whole range is ±0.01, so every point on it is already inside the band.

The two LCMAP panels should be read as noise, not signal — including their excursion above zero. A four-seed repeat (page 5) puts their split noise at ±0.023, wider than the features visible here. LUCAS, five times larger, is the panel to trust.

Why cosine error cannot be compared across sources

The same plot against reconstruction error instead of rate. If cosine error predicted downstream harm, the two curves would lie on top of each other. They do not — Tessera tolerates far more of it.



Blue (AlphaEarth) occupies the low-error end of the axis, red (Tessera) the high-error end — they barely overlap. Where they do, the ordering is the opposite of what a cosine-based intuition predicts:

LUCAS LC AE @ cos 0.9532 Δ −0.043
Tess @ cos 0.9475 Δ −0.005

ASTER GED AE @ cos 0.9767 Δ −0.001
Tess @ cos 0.9640 Δ +0.007

At an equal or slightly worse reconstruction cosine, the Tessera codec costs less downstream accuracy than the AlphaEarth codec. On LUCAS the 0.038 gap is several times the ±0.011 split noise; ASTER's is small but both sit at zero, which makes the same point.

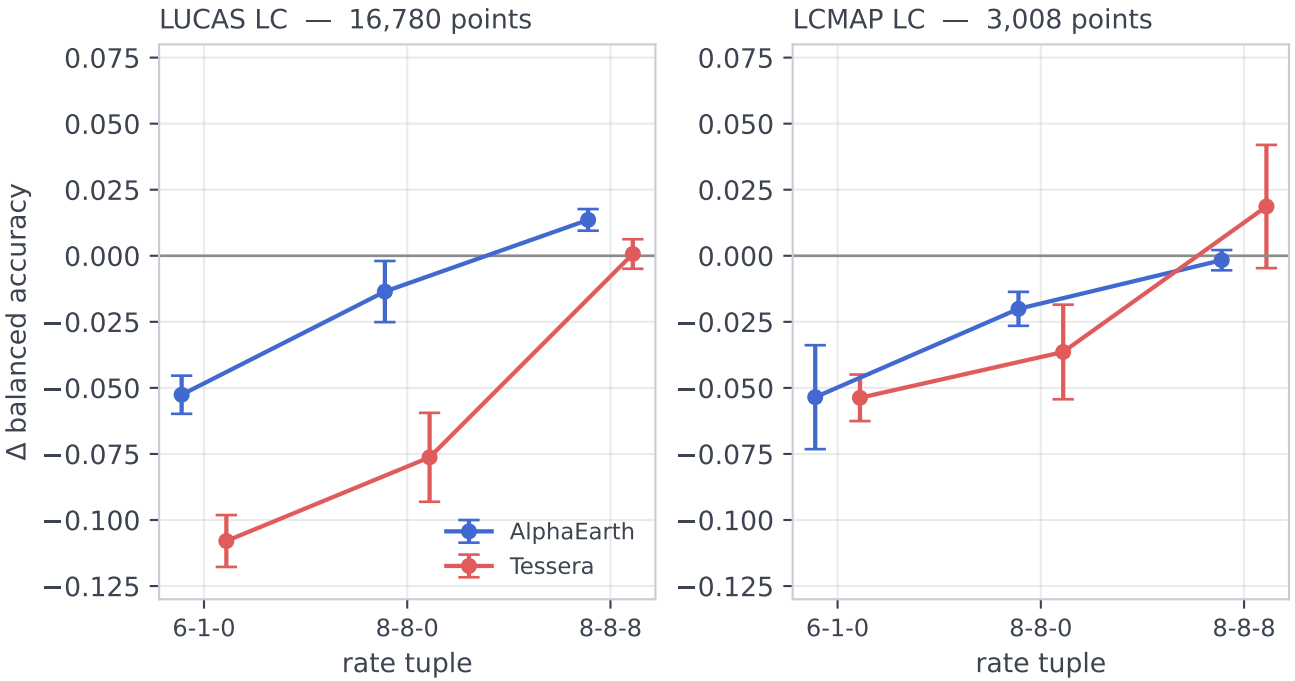
The interpretation is that the two sources spend their vector norm differently. Tessera's embeddings carry ~2.5x more independent variation per pixel (PCA d95: 51/91/71 of 512/512/256 tier dims, against AlphaEarth's 19/34/28), much of it fine-scale and weakly correlated with land cover, land use or emissivity. Discarding it costs cosine but not task accuracy. AlphaEarth's field is smoother and more nearly low-rank, so a given cosine error necessarily eats into the dimensions the probes actually read.

Practical consequence: a cosine target transplanted from the AlphaEarth codec (0.998) is the wrong specification for Tessera. Set Tessera's quality target from downstream transparency — which this run reaches around 1 b/px at cos ~0.93 — not from AlphaEarth's error budget.

Only the LUCAS and ASTER rows above carry weight: the LCMAP points, plotted faintly here for completeness, sit inside a ±0.05 noise floor (page 5) and cannot order the two sources either way. The LUCAS comparison — 16,780 points, ±0.011 noise — is the evidence.

How much of this is noise?

Every rate point re-probed under repeated geo-block split seeds (n per row in the table). The error bars are what a single-seed curve cannot s



What the repeat shows

LUCAS (left) behaves: the seed spread is ±0.005-0.011, the two sources separate cleanly at low rate, and both converge to zero at 8-8-8. The low-rate gap between red and blue is 4-9x the noise, so it is real.

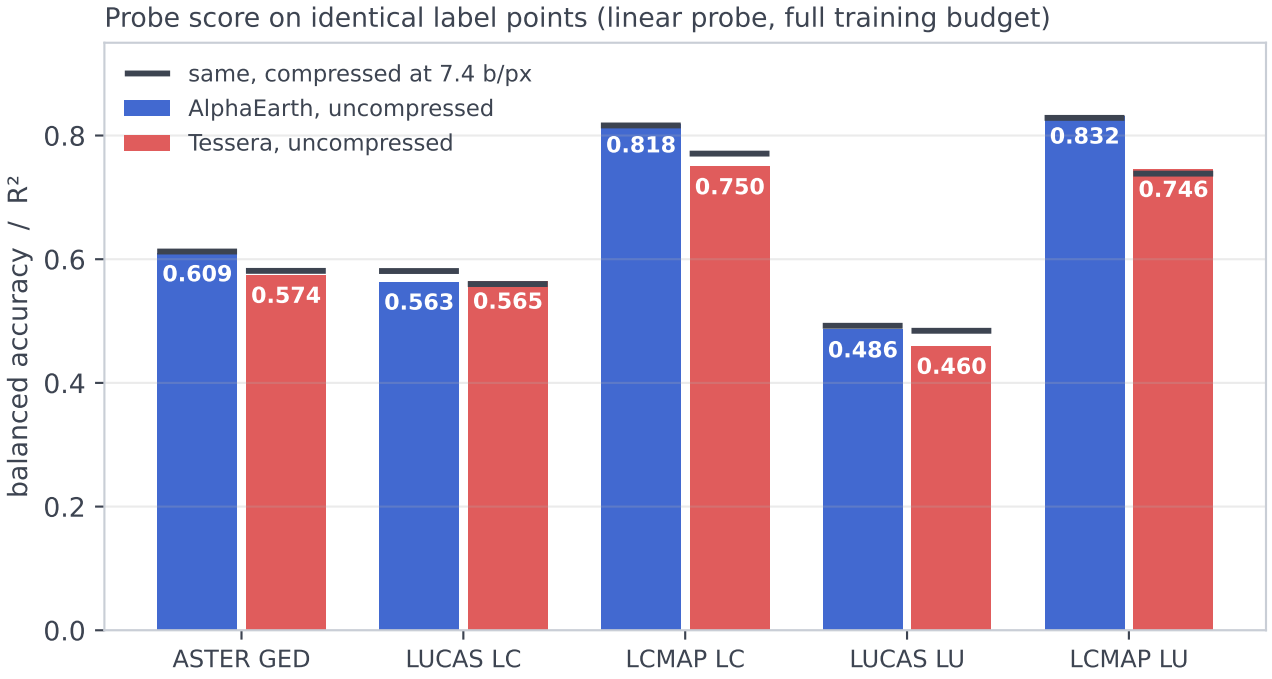
LCMAP (right) does not. Its seed spread reaches ±0.023 — wider than most of the effects being discussed. The +0.02 "compression beats the original" reading on page 3 disappears into it: across four seeds the same configuration returns -0.004, +0.002, +0.021 and +0.056.

The lesson is not that LCMAP disagrees. It is that a single split on 3,008 points cannot answer this question at all, and a one-seed curve gives no hint of that.

benchmark	source	rate	mean Δ	sd	n
LUCAS LC	AlphaEarth	6-1-0	-0.053	0.007	3
LUCAS LC	AlphaEarth	8-8-0	-0.014	0.012	3
LUCAS LC	AlphaEarth	8-8-8	0.014	0.004	3
LUCAS LC	Tessera	6-1-0	-0.108	0.010	3
LUCAS LC	Tessera	8-8-0	-0.076	0.017	3
LUCAS LC	Tessera	8-8-8	0.001	0.006	3
LCMAP LC	AlphaEarth	6-1-0	-0.054	0.020	4
LCMAP LC	AlphaEarth	8-8-0	-0.020	0.006	4
LCMAP LC	AlphaEarth	8-8-8	-0.002	0.004	4
LCMAP LC	Tessera	6-1-0	-0.054	0.009	4
LCMAP LC	Tessera	8-8-0	-0.036	0.018	4
LCMAP LC	Tessera	8-8-8	0.019	0.023	4

Absolute performance: the sources themselves

Compression cost is only half the question. On the same points, how good are the two embedding fields to begin with?



Dark ticks mark each source at the top of its rate ladder — compressed scores sit essentially on top of the uncompressed bars, which is the transparency result restated.

AlphaEarth leads on four of the five tasks, by 3-7 points of balanced accuracy on the LCMAP pair and 0.035 of R^2 on ASTER.

The exception is LUCAS land cover, where the two are level (0.5635 vs 0.5650).

This is a statement about these five probe tasks, not about the sources in general: all five are coarse thematic labels at a single point, the regime AlphaEarth was designed and tuned for. Tessera's extra per-pixel information may well pay off on tasks with finer spatial or temporal structure — none of which are in this benchmark suite.

The numbers behind pages 3-5 (linear probe, full training budget, identical points)

	uncompressed		Δ at 0.07 b/px		Δ at 0.43 b/px		Δ at 7.4 b/px	
benchmark	**AE**	**Tess**	**AE**	**Tess**	**AE**	**Tess**	**AE**	**Tess**
ASTER GED	0.609	0.574	-0.001	-0.004	-0.002	0.011	0.003	0.007
LUCAS LC	0.563	0.565	-0.043	-0.098	0.003	-0.059	0.017	-0.005
LCMAP LC	0.818	0.750	-0.026	-0.059	-0.023	-0.023	-0.001	0.021
LUCAS LU	0.486	0.460	-0.021	-0.028	-0.017	-0.010	0.006	0.025
LCMAP LU	0.832	0.746	-0.052	-0.066	-0.037	-0.044	-0.003	-0.007

ASTER GED is R^2; the other four are balanced accuracy. Negative Δ = compression costs accuracy.

Caveats and what to do next

Caveats

- The codec is interim. tessera_vq8 stopped at 47k of 60k steps (validation cosine 0.962 against the 0.980 stage-1 ceiling the architecture reaches). A finished codec should move the whole Tessera curve left; the low-rate gap is the part most likely to shrink.
- LCMAP cannot resolve this question. 3,008 points give a ±0.023 seed-to-seed standard deviation — wider than most of the effects discussed here. Its panels are shown for completeness only, and no claim in this report rests on them. ASTER has not been seed-checked; treat it likewise until it has.
- Rate is not bytes. Bits per pixel here is the nominal RVQ index cost. Tessera carries 128 bands to AlphaEarth's 64, so at equal bits/pixel Tessera is compressing twice as much data — its compression factor at 1 b/px is roughly double AlphaEarth's at the same rate.
- Sub-sampled coverage. LUCAS and LCMAP were capped at 2,000 Tessera tiles, so these are ~17% and ~25% subsets of the AlphaEarth study's point sets (both sources probed on the subset).
- Five coarse thematic tasks. Nothing here tests temporal, sub-pixel or fine-texture use of the embeddings, where the sources differ most.

Next steps, in order of value

1. Re-run this suite against the final Tessera codec when stage-2 finishes. The scripts take a checkpoint path; the extracted datasets are reusable, so a rerun is minutes of GPU, not hours.
2. Set the Tessera quality target from the transparency knee (~1 b/px on this run), not from AlphaEarth's cosine spec. If that knee holds for the final codec it is the number that belongs in the release contract.
3. Add a task that exercises what Tessera has and AlphaEarth does not — fine-scale or multi-temporal structure. The current suite cannot see Tessera's advantage, only its cost.
4. Raise LCMAP's tile budget (it is 25% sampled) or retire it. As it stands it costs GPU time and returns noise. Seed-check ASTER at the same time.

Reproducing

Datasets /data/tessera-samples/eval/{aster_ged,lucas_lc,lcmap_lc}
Results ring-space/plots/tessera_downstream/
(Tessera sweeps, ae_matched/, seedcheck/, seedcheck_lucas/)
Code branch worktree-tessera-source — georvq/eval/tessera_extract.py, scripts/eval_build_dataset_tessera.py, scripts/eval_rate_sweep.py --restrict-parquet, scripts/tessera_downstream_report.py (this document)